



IILM UNIVERSITY GREATER
NOIDA, UTTAR PRADESH

📖 INTERNSHIP PRESENTATION

DATA ANALYTICS

An Internship Report on Data Analytics

Submitted to School of Computer Science and Engineering in partial fulfilment of the requirement for the degree of Bachelor of Technology in Computer Science and Engineering.

🎓 Academic Batch: 2024 – 2028

August
2026

📖 CANDIDATE CREDENTIALS

STUDENT NAME

TANISHQ RAWAT

ROLL NUMBER

210030475

DEGREE & DEPARTMENT

B.Tech CSE

ASSESSMENT & PLATFORM

LearnTube.ai

Backed by Google for Startups

ID: DJA-B-1-2999033-0

Certification Successfully Completed



Declaration & Acknowledgement

Candidate's Declaration

I, **TANISHQ RAWAT** (Roll No. **210030475**), hereby declare that the internship report titled "**DATA ANALYTICS**" has been completed to fulfill the degree requirement for Bachelor of Technology in Computer Science and Engineering at IILM University.
I affirm that this work is original, all references have been cited appropriately, and it has not been submitted for any other degree or diploma program.

Acknowledgements

I express my sincere gratitude to the faculty of the **School of Computer Science and Engineering**, IILM University, Greater Noida, for their continuous support and academic guidance. Special thanks to my industry mentors for providing hands-on opportunities on real-world data analytics problems, and to my parents for their constant blessings.

Organization Profile

Data-Driven Solutions

The internship was conducted at an organization providing end-to-end data analytics solutions across key industry verticals including retail, finance, and healthcare.

Core Competencies

Focuses on converting raw, unstructured business data into actionable strategic insights using statistical modeling, machine learning algorithms, and interactive BI dashboards.

Collaborative Work Culture

Operates in an agile environment promoting peer code reviews, continuous data pipeline integration, and direct mentorship from experienced data analysts.



Problem Statement

80%

UNSTRUCTURED DATA

10x

DATA VOLUME GROWTH

< 20%

DATA UTILIZATION

100%

DECISION RELIANCE

The Industry Challenge

Modern enterprises generate vast volumes of transactional and operational data daily. However, a significant portion remains under-utilized due to fragmented processing systems and the absence of structured analytical frameworks.

Project Objective: Design and implement an end-to-end data analytics workflow capable of cleaning, processing, and analyzing real-world datasets to uncover underlying trends, detect anomalies, and empower stakeholders with predictive insights.

Project Objectives



Data Consolidation Collect and aggregate raw data from diverse sources into a single, structured dataset.



Data Preprocessing Clean and standardize data by handling missing records, duplicates, and structural anomalies.



Exploratory Analysis (EDA) Perform comprehensive statistical checks to identify key patterns, correlations, and outliers.



Interactive Dashboards Design visually compelling Power BI / Tableau dashboards for strategic visualization.



Predictive Modeling Apply statistical machine learning algorithms to generate churn forecasts and trend metrics.



Actionable Reporting Document key analytical findings and communicate data-driven recommendations to leadership.

Scope of the Project

✓ In-Scope Activities

- ✓ End-to-end analysis of organizational domain dataset over the fixed internship duration.
- ✓ Data ingestion, quality checks, missing value imputation, and feature engineering.
- ✓ Development of predictive modeling prototypes (regression/classification).
- ✓ Creation of executive dashboards for business intelligence communication.

⊘ Out-of-Scope & Future Work

- ✗ Full-scale microservice deployment into live production infrastructure.
- ✗ Real-time streaming data ingestion pipelines (Kafka / Spark Streaming).
- 💡 **Future Scope:** Automating data pipeline ingestion and model re-training triggers.

Technologies & Tools

Category	Tool / Technology	Application & Purpose
Programming	Python, Pandas, NumPy	Data manipulation, array operations, and data frame transformation
ML & Viz Libraries	Scikit-learn, Matplotlib, Seaborn	Predictive modeling, statistical plotting, correlation heatmaps
Querying	SQL	Data extraction, join operations, and database aggregations
BI & Viz Tools	Power BI, Tableau	Building interactive business dashboards and KPI reports
Environments	Jupyter Notebook, VS Code	Interactive EDA prototyping and modular Python scripting
Version Control	Git, GitHub, MS Excel	Code versioning, collaboration, and preliminary data inspection

System Architecture



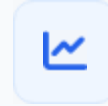
1. Data Sources

Ingestion of raw operational data from SQL databases, REST APIs, CSV files, and enterprise spreadsheets.



2. Processing Layer

Data cleaning, outlier filtering, normalization, missing value handling, and feature extraction using Python.



3. Analytics Layer

Applying exploratory statistical analysis, correlation matrix checks, and machine learning models (Scikit-learn).



4. Presentation

Delivering interactive dashboards, KPI visualization metrics, and analytical reports for leadership decisions.

Methodology & Workflow

STAGE1

Collection

Gathering raw datasets from host organization databases.

STAGE2

Cleaning

Handling missing values, deduplication, format alignment.

STAGE3

EDA

Summary statistics, distributions, outlier detection.

STAGE4

Engineering

Creating domain variables to boost model accuracy.

STAGE5

Modeling

Applying regression & classification algorithms.

STAGE6

Evaluation

Evaluating RMSE, Accuracy, Precision metrics.

STAGE7

Reporting

Building dashboards & stakeholder presentations.

Expected Outcomes



Cleaned Standardized Dataset Delivered a reliable, fully preprocessed dataset ready for statistical modeling and downstream BI tools.



Visual Pattern Discovery Uncovered critical business trends, customer behavior metrics, and operational bottlenecks.



Interactive BI Dashboards Empowered non-technical business stakeholders to slice, filter, and analyze operational metrics independently.



Predictive Analytics Insights Trained predictive models to forecast customer churn and demand under varying business conditions.

Certification & Performance

Certificate Verification

VERIFIED

LearnTube.ai Assessment

Backed by Google for Startups

Certificate ID: DJA-B-1-2999033-0
Issue Date: August 30, 2026
Issued By: Shronit Ladhani & Gargi Ruparelia

Assessment covered EDA, churn metrics, predictive modeling, and business forecasting.

100%

Perfect Assessment Score

10.0 / 10.0 correct answers, placing in the elite top 5% of assessed data professionals.

**#3
9**

Global & Weekly Rank

Ranked #39 globally out of 74,307 participants and #39 out of 154 weekly candidates.

Demonstrated Key Competencies

Strong problem-solving, strategic metric selection, churn prediction, and clear communication of analytical insights.

Bibliography & References

 **[1] Wes McKinney, "Python for Data Analysis"** O'Reilly Media, 3rd Edition, 2022. Covers core Pandas, NumPy, and data manipulation patterns.

 **[2] Jiawei Han, Micheline Kamber, Jian Pei, "Data Mining: Concepts and Techniques"** Morgan Kaufmann, 3rd Edition. Fundamental data preprocessing and statistical clustering concepts.

 **[3] Official Documentation Frameworks** Pandas Documentation (pandas.pydata.org), Scikit-Learn Guide (scikit-learn.org), and Microsoft Power BI Documentation.

 **[4] LearnTube.ai Assessment & Internal Training Material** Data Analytics Assessment Certificate DJA-B-1-2999033-0 and host organization internal project guidelines.

Thank You!

Questions & Discussion

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Program **B.Tech CSE
(2024-2028)**

Institution **IILM University,
Greater Noida**

Image Sources



<https://www.rib-software.com/app/uploads/2024/06/dashboard-design-principles-blog-rib.webp>

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